

FORT GRATIOT LH, MI, USA

WMO: 997258

Lat: 43.0063N

Lon: 82.4224W

Elev: 180

StdP: 99.18

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.3	(c) -14.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.5	(k) -4.8	(l) 12.1	(m) -2.8	(n) 4.7	(o) 220	(p) 0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.9	(c) 31.1	(d) N/A	(e) 29.4	(f) N/A	(g) 27.7	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.7	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.6	(b) 11.2	(c) 9.3	(e) -20.0	(f) 34.2	(g) 3.9	(h) 2.1	(i) -22.8	(j) 35.7	(k) -25.0	(l) 37.0	(m) -27.2	(n) 38.2	(o) -30.0	(p) 39.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-4.7	-4.5	0.7	7.0	13.3	18.4	21.5	21.0	17.6	11.2	4.4	-0.2		
	DBStd	10.36	5.91	5.51	5.55	4.85	4.87	4.20	3.45	2.98	3.83	4.36	4.80	4.66		
	HDD10.0	1821	455	406	293	117	19	1	0	0	1	35	176	318		
	HDD18.3	3753	713	639	546	339	171	51	10	9	59	224	417	576		
	CDD10.0	1417	1	0	6	28	122	253	357	341	227	74	9	1		
	CDD18.3	308	0	0	0	0	15	53	108	91	36	4	0	0		
	CDH23.3	1996	0	0	1	14	133	394	736	499	189	29	0	0		
	CDH26.7	568	0	0	0	2	32	123	233	122	48	9	0	0		
(14) Wind	WSAvg	4.6	4.8	4.9	5.2	5.4	4.5	4.2	3.9	3.9	4.1	4.7	4.7	4.7		
(15) Precipitation	PrecAvg	866	48	52	67	75	68	80	69	77	92	64	86	73		
	PrecMax	1102	85	109	146	132	123	135	108	180	251	119	134	117		
	PrecMin	640	24	15	30	39	23	21	31	21	12	17	17	33		
	PrecStd	115	20	29	27	23	27	27	25	50	61	29	35	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	11.8	22.0	25.5	30.0	32.8	34.3	31.7	31.3	26.7	19.0	14.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	8.9	8.5	16.8	22.1	27.4	30.3	31.4	29.7	27.7	22.4	15.5	10.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	6.1	5.0	12.8	18.7	24.7	27.7	29.2	28.0	25.4	20.1	13.6	8.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	3.4	2.7	8.9	15.5	21.9	25.3	27.4	26.2	23.5	17.9	11.7	6.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	6.8	7.4	8.2	8.8	8.3	7.9	7.6	8.0	7.6	6.5	5.5		
		MCDBR	8.3	9.3	12.9	14.7	13.2	11.8	11.0	10.0	10.2	9.8	9.6	8.1		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.320	0.338	0.383	0.408	0.432	0.420	0.409	0.377	0.358	0.319	0.294		
	taud		2.338	2.302	2.371	2.275	2.261	2.222	2.282	2.300	2.376	2.405	2.485	2.445		
	Ebn at Noon		818	872	904	882	867	843	850	847	850	820	810	810		
	Edh at Noon		82	101	107	128	133	139	130	123	106	91	72	68		
(44) All-Sky Solar Radiation	RadAvg		1.31	2.15	3.45	4.55	5.36	6.08	6.00	5.21	4.17	2.66	1.59	1.08		
	RadStd		0.17	0.22	0.35	0.40	0.46	0.34	0.32	0.24	0.37	0.23	0.20	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page